provided responses to CDP's questionnaire. Approximately 610 of these were Commission registrants,[2682] suggesting that 10 percent of the approximately 5,860 domestic registrants that will be subject to the final rules provided responses to CDP's questionnaire in 2022. The response rate was higher among companies with higher market capitalizations. For example, CDP lists 351 respondents as included in its S&P 500 sample, suggesting that approximately 70 percent of S&P 500 companies provided responses. Of these 351 respondents, 95 percent provided Scope 1 and Scope 2 emissions data. In addition, a 2022 report examines Russell 1000 companies and finds that 57% disclose Scope 1 and 2 emissions.[2683]

c. Use of Third-Party Frameworks

Multiple third-party reporting frameworks and data providers have emerged over the years to facilitate and encourage the reporting of climate-related information by companies.[2684]

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impact and performance. To aid comparability and ensure comprehensiveness, CDP includes sector-specific questions and data points. In 2018, CDP aligned its climate change questionnaire with the TCFD. Companies' participation in the CDP questionnaire is voluntary. If a company decides to respond to the questionnaire and disclose its information to the CDP, it then has the option to mark its response as either "Public" or "Private." Importantly, responses marked as "Private" are available only to the signatory investors of the CDP (non-signatory investors and the general public cannot access this information). Responses marked as "Public" can be accessed by the general public at no cost. *See* CDP, available at https://www.cdp.net/en/info/about-us. In a meeting with CDP officials, the Commission staff was informed that the number of public companies that respond to the CDP questionnaire but do not publicly disclose their responses is negligible. *See* SEC Meeting Memorandum dated June 15, 2023, available at https://www.sec.gov/comments/s7-10-22/s71022-206619-416182.pdf.

[2682] This estimate is based on matching CDP survey respondents to registrants on ticker, company name, and industry. Five-hundred seventy matches were made on ticker. Approximately 40 more matches were made on company name using Levenshtein Distance. The matches were then manually reviewed by Commission staff to ensure the industry description provided by CDP aligned with the SIC code assigned to the matched registrant.

[2683] *See* Just Capital, The Current State of Environment Disclosure in Corporate America: Assessing What Data Russell 1000 Companies Publicly Share, available at https://justcapital.com/wp-content/uploads/2022/04/JUST-Capital_Environment-State-of-Disclosure-Report_2022.pdf.

[2684] The TCFD, the SASB, the GRI, the Principles for Responsible Investment, the PCAF, and the CDP (among others), have all developed standards and systems that aim to help firms and investors identify, measure, and communicate climate-related information and incorporate that information into their business practices. Multiple frameworks have emerged, in part, because each seeks to provide different information or fulfill different functions when it comes to disclosing information related to climate-related risks or other ESG factors that may be important to investors.

Due to the voluntary nature of third-party frameworks, however, companies often disclose some but not all components of those frameworks, and the components that are disclosed may not be the same across companies,[2685] resulting in reporting fragmentation.

Some companies follow existing third-party reporting frameworks when developing climate-related disclosures for Commission filings or to be included in CSR, sustainability, ESG, or similar reports. As described in the Proposing Release, for instance, one survey found that 59 percent of respondents follow one or more such frameworks.[2686] Among these respondents, 44 percent used SASB, 31 percent used the Global Reporting Initiative GRI, 29 percent used the TCFD, and 24 percent used the CDP.[2687] Broadly similar statistics on the usage of different reporting frameworks are also provided by other studies. For example, another report[2688] found that 78 percent of sustainability reports from Russell 1000 companies aligned with SASB reporting standards,[2689] 54 percent utilized GRI reporting standards,[2690] 50 percent aligned with the TCFD recommendations,[2691] and 53 percent responded to the CDP Climate Change

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2685 *See* Reiners *et al.*, *supra* note 2653.

2686 *See* s*upra* note 2662; *see also* Proposing Release, section IV.A.5.

2687 *See* Proposing Release at nn.768-770 and accompanying text.

2688 *See supra* note 2666.

2689 *See id.* Sixty-seven percent of companies in the smaller half of the Russell 1000 index (by market capitalization) report according to SASB standards. The corresponding statistic for companies in the larger half is 88%.

2690 *See id.* Forty percent of companies in the smaller half of the Russell 1000 index (by market capitalization) report according to GRI standards. The corresponding statistic for companies in the larger half is 66%.

2691 *See id.* Thirty-two percent of companies in the smaller half of the Russell 1000 index (by market capitalization) report according to the TCFD recommendations. The corresponding statistic for companies in the larger half is 65%.

questionnaire.[2692] A review of website sustainability disclosures by 80 small- and mid-cap companies across five different industries found comparable numbers.[2693]

While these various frameworks are distinct, they overlap in their alignment with the TCFD recommendations. According to one report,[2694] the GRI standards exhibit "Reasonable" alignment with the TCFD, while the SASB standards generally exhibit "Moderate" or "Reasonable" alignment with the majority of the TCFD disclosure items. Additionally, the CDP Climate Change questionnaire fully incorporates the TCFD framework and thus exhibits full alignment.[2695] Thus, companies that report following the GRI, SASB, or CDP frameworks are, to varying degrees, producing disclosures that are in line with the TCFD. However, because each framework takes different approaches (*e.g.*, different intended audience and/or reporting channel) and because certain differences exist in the scope and definitions of certain elements, investors may find it difficult to compare disclosures under each framework. One organization

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2692 *See id.* Thirty-two percent of companies in the smaller half of the Russell 1000 index (by market capitalization) responded to the CDP Climate Change questionnaire. The corresponding statistic for companies in the larger half is 74%.

2693 *See* White & Case and the Soc. Corp. Gov., *A Survey and In-Depth Review of Sustainability Disclosures by Small- and Mid-Cap Companies* (Feb. 18, 2021), available at https://www.whitecase.com/publications/article/survey-and-depth-review-sustainability-disclosures-small-and-mid-cap-companies (Among the companies reviewed, 41 companies (51%) provided some form of voluntary sustainability disclosure on their websites. Further, nine of those 41 companies indicated the reporting standards with which they aligned their reporting, with the majority of the nine companies not following any one set of standards completely. Additionally, six companies followed the GRI standards, while three companies stated that they follow both the TCFD recommendations and SASB standards).

2694 The Corporate Reporting Dialogue is a platform, convened by the Value Reporting Foundation, to promote greater coherence, consistency and comparability between corporate reporting frameworks, standards and related requirements. *See Driving Alignment in Climate-related Reporting, Corporate Reporting Dialogue* (2019), available at https://www.integratedreporting.org/wp-content/uploads/2019/09/CRD_BAP_Report_2019.pdf (providing a detailed assessment of the various frameworks' degrees of alignment with each TCFD disclosure item, ranging from maximum to minimum alignment as follows: Full, Reasonable, Moderate, Very Limited, and None).

2695 *See* CDP, *supra* note 52.

analyzed the rate of disclosure for each TCFD disclosure element for a sample of 659 U.S. companies in 2020 and 2021, presented in Table 6.[2696]

Table 6. Third-party analysis of TCFD disclosure rates from a sample of U.S. companies[1]

TCFD Disclosure Element		Rate of Disclosure
Governance	a) Describe the board's oversight of climate-related risks and opportunities.	17%
	b) Describe management's role in assessing and managing climate-related risks and opportunities.	10%
Strategy	a) Describe the climate-related risks and opportunities the organization has identified over the short, medium, and long term.	45%
	b) Describe the impact of climate-related risks and opportunities on the organization's businesses, strategy, and financial planning.	34%
	c) Describe the resilience of the organization's strategy, taking into consideration different climate-related scenarios, including a 2°C or lower scenario.	5%
Risk Management	a) Describe the organization's processes for identifying and assessing climate-related risks.	15%
	b) Describe the organization's processes for managing climate-related risks.	17%
	c) Describe how processes for identifying, assessing, and managing climate-related risks are integrated into the organization's overall risk management.	16%
Metrics and Targets	a) Describe the metrics used by the organization to assess climate-related risks and opportunities in line with its strategy and risk management process.	21%

[2696] *See* Moody's Analytics, TCFD-Aligned Reporting by Major U.S. and European Corporations (Feb. 2022), available at https://www.moodysanalytics.com/articles/pa/2022/tcfd_aligned_reporting_by_major_us_and_european_corporations. The sample for analysis was provided to Moody's Analytics by the TCFD and includes 659 companies domiciled in the United States. To arrive at these statistics, Moody's conducted an artificial intelligence ("AI") based review of all public filings, including financial filings, annual reports, integrated reports, sustainability reports, and other publicly available reports that were associated with companies' annual reporting on sustainability. Non-public disclosures, such as responses to the CDP questionnaire, were not included in the analysis.

	b) Disclose Scope 1, Scope 2, and, if appropriate, Scope 3 GHG emissions, and the related risks.	19%
	c) Describe the targets used by the organization to manage climate-related risks and opportunities and performance against targets.	25%

1. The source of this table is Moody's Analytics. *See* supra note 2696.

The variety of disclosure frameworks in use, and their varying rates of overlap with the TCFD disclosure elements, demonstrates the low rate of consistency and comparability among existing climate disclosures.

d. Climate-Related Targets, Goals, and Transition Plan Disclosures

Carbon reduction targets or goals have become an increasing focus for both companies and countries.[2697] For example, 195 parties, including the United States, the EU, and the UK, have signed the Paris Climate Agreement as of December 2023.[2698] The agreement aims to strengthen the global response to climate change by keeping a rise in global temperatures to well below 2º Celsius above pre-industrial levels this century, as well as pursue efforts to limit the temperature increase even further to 1.5º Celsius.[2699] A 2022 report, which examined approximately 5,300 companies across the globe, found that over one-third of these companies

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2697 *See, e.g.,* UNFCC COP28 Agreement, *supra* note 34; Press Release, United Nations Framework Convention on Climate Change, Commitments to Net Zero Double in Less Than a Year, (Sept. 21, 2020), available at https://unfccc.int/news/commitments-to-net-zero-double-in-less-than-a-year.

2698 *See* United Nations, Multilateral Treaties Deposited with the Secretary General ch. XXVII, 7.d Paris Agreement (treaty status updated Feb. 2024), available at https://treaties.un.org/Pages/ViewDetails.aspx?src=TREATY&mtdsg_no=XXVII-7-d&chapter=27&clang=_en treaty collection ; *see also* EU Press Release, *Corporate Sustainability Due Diligence: Council and Parliament Strike a Deal to Protect Environment and Human Rights* (Dec. 14, 2023), available at https://www.consilium.europa.eu/en/press/press-releases/2023/12/14/corporate-sustainability-due-diligence-council-and-parliament-strike-deal-to-protect-environment-and-human-rights/ (announcing a provisional agreement to adopt the Corporate Sustainability Due Diligence Directive, which includes a requirement that companies ensure their business strategies are compatible with limiting global warming to 1.5º Celsius).

2699 *See* section I.

announced plans to curb their Scope 1 or Scope 2 emissions.[2700] Of these 5,300 companies that also responded to the CDP climate survey, the same report found that about one-fourth of these companies had established a target to achieve net-zero carbon emissions.[2701] In addition, a growing number of companies and organizations have signed on to The Climate Pledge, indicating a commitment to achieve net-zero emissions by 2040.[2702] According to data from another source, as of August 2023, 5,728 companies had established climate targets.[2703] Of these companies, 710 were located in the United States, about half of which were Commission registrants.[2704] The trend in companies disclosing other climate-related targets has also been increasing over time.[2705]

An increasing number of companies are adopting transition plans, according to a 2023 report.[2706] This report finds that 4,100 organizations[2707] across the globe reported having transition plans aligned with reaching a temperature change of no more than 1.5º Celsius above

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2700 *The Sustainability Yearbook 2022*, S&P Global (Feb. 2022), available at https://www.spglobal.com/esg/csa/yearbook/2022/downloads/spglobal_sustainability_yearbook_2022.pdf.

2701 *Id.*

2702 As of February 20, 2024, The Climate Pledge had acquired 468 signatories, 146 of which are from the United States. *See The Climate Pledge*, available at https://www.theclimatepledge.com/us/en/Signatories (last visited Feb. 20, 2024).

2703 *See Target Dashboard*, available at https://sciencebasedtargets.org/ (as visited Aug. 16, 2023).

2704 *See id.*

2705 For example, the percentage of both global and U.S. companies with water reduction targets grew by 4% in 2019 on a year-over-year basis. This represented 28% of major global companies (i.e., those listed on the S&P Global 1200 index) and 27% of major (i.e., those listed in the S&P 500 index) U.S. companies publicly disclosing these targets. *See State of Green Business 2021*, S&P Global (Feb. 4, 2021), available at https://www.spglobal.com/marketintelligence/en/news-insights/research/state-of-green-business-2021.

2706 CDP, Are Companies Developing Credible Climate Transition Plans? (Feb. 2023), available at https://cdn.cdp.net/cdp-production/cms/reports/documents/000/006/785/original/Climate_transition_plan_report_2022_%2810%29.pdf.

2707 *See id.* According to the CDP's Transition Plan report, "Nearly 20,000 organizations around the world disclosed data through CDP in 2022, including more than 18,700 companies worth 50% of global market capitalization, and over 1,100 cities, states and regions."

pre-industrial levels.[2708] Approximately 43 percent of these transition plans are publicly available.[2709]

Commission staff compared these figures to data related to targets and goals on the Bloomberg ESG database, which is focused on registrants listed on NYSE and NASDAQ. The results are reported in Table 7 below. These results are generally consistent with data from the sources discussed above.

[2708] *See id.*

[2709] *See id.*

Table 7. Registrants with targets or goals according to Bloomberg ESG data

	Climate Change Policy[1]	Emission Reduction Initiatives[2]	Science-Based Targets[3]	Net Zero Plans[4]
All issuers	37%	45%	11%	17%
NAFs	10%	11%	1%	2%
AFs	23%	29%	2%	7%
LAFs	55%	67%	19%	27%
EGCs	8%	9%	0%	1%
Non EGCs	47%	58%	15%	22%
SRCs	13%	12%	0%	3%
Non SRCs	40%	49%	12%	18%

Sources: Bloomberg, SEC filings

1 Bloomberg defines this field as indicating: "Whether the registrant has disclosed its intention to help reduce global GHG emissions through its ongoing operations and/or the use of its products and services in its annual report or CSR report. Examples might include efforts to reduce GHG emissions, efforts to improve energy efficiency, efforts to derive energy from cleaner fuel sources, investment in product development to reduce emissions generated or energy consumed in the use of the company's products etc."

2 Bloomberg defines this field as indicating: "Whether the registrant has disclosed the implementation of any initiative to reduce its emissions, such as GHGs, SOx, NOx, or other air pollutants in its annual report or CSR report."

3 Bloomberg defines this field as indicating: "Whether the registrant has disclosed its ambition and engagement related to setting science-based GHG emissions reduction targets. Emissions targets are considered science-based if they align with the goals of the Paris Climate Agreement to limit warming to well below 2 degrees Celsius above pre-industrial levels. That is, whether the company has explicitly disclosed that they have either committed to setting or have set science-based targets. This information is sourced from a company's CSR report."

4 Bloomberg defines this field as indicating: "Whether the registrant has disclosed its ambition and engagement related to achieving Net Zero GHG emissions. Net Zero refers to a state in which GHG emissions released into the atmosphere are balanced by removal of emissions from the atmosphere. This information is sourced from a company's CSR report."

The results suggest that smaller registrants (NAFs, EGCs, and SRCs) are much less likely to have developed climate-related targets and goals. For example, the portion of companies that have "Net Zero Plans" is approximately 1 percent for EGCs and approximately 22 percent for non-EGCs.

e. Third-Party Assurance of Climate-Related Disclosures

Among the companies that provide climate-related disclosures, a considerable portion include some form of third-party assurance of the accuracy of these disclosures. One report finds that 40 percent of Russell 1000 Index companies, nearly all of which are LAFs, obtained third-party assurance for their sustainability reports in 2022, up from 24 percent in 2019.[2710] Among the companies that obtained assurance, however, only three percent obtained assurance for the entire report, with 58 percent obtaining assurance only with respect to GHG emissions. Regarding the level of assurance, the overwhelming majority (92 percent) obtained limited assurance while only 5 percent obtained reasonable assurance. Regarding service providers, 17 percent of companies received assurance from an accounting firm, 15 percent from small consultancy/boutique firms, and 68 percent from engineering firms.[2711] Because these statistics are limited to Russell 1000 Index companies, corresponding figures for the full sample of U.S. registrants may differ depending on the extent to which the practice of obtaining third-party assurance is concentrated in large companies.[2712] Indeed, based on Commission staff's analysis

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2710 *See supra* note 2666.

2711 One study finds that assurance service providers that are not financial auditors are reported to be not applying the AICPA assurance standards. *See* Brandon Gipper, Samantha Ross & Shawn Shi, *ESG Assurance in the United States* (Aug. 14, 2023), Stanford Univ. Grad. Sch. of Bus. Rsch Paper No. 4263085, UC San Francisco Rsch. Paper No. Forthcoming, available at https://ssrn.com/abstract=4263085 (retrieved from SSRN Elsevier database) ("Gipper *et al*. (2023)"); *see also supra* note 1363 (explaining that non-CPAs are unable to use AICPA or PCAOB attestation standards).

2712 Other studies also report evidence of third-party assurance among smaller samples of companies analyzed. For example, according to a recent study by the International Federation of Accountants ("IFAC"), in 2019, 99 out of the 100 largest U.S. companies by market capitalization provided some form of sustainability disclosure, which may contain climate-related information among other sustainability-related topics. Seventy of those companies obtained some level of third-party assurance, with the vast majority being "limited assurance" according to the study. Of the 70 companies that obtained assurance, the study reports that 54 obtained "limited assurance," eight obtained "reasonable assurance," five obtained "moderate assurance," and three did not disclose any assurance. Of the 81 unique assurance reports examined in the study, nine were found to be issued by an auditing firm, while 72 were issued by another service provider. *See* IFAC, *supra* note 1089. Among the sample of 436 companies included in the CCMC Survey, 28% disclosed that they engaged a third party to provide some form of auditing or assurance regarding their climate-related or ESG disclosure.

of Bloomberg ESG data, which focuses on registrants listed on NYSE and NASDAQ, approximately 15 percent obtained some type of third-party assurance or verification[2713] on their environmental policies and data, nearly all of which are non-SRCs and non-EGCs.[2714] Based on analysis of S&P 500 companies from 2010 through 2020, a 2023 study finds that the most common form of assurance standard used for GHG emissions is the ISO 14064,[2715] which is the assurance standard typically applied by assurance providers who are not accountants.[2716] Specifically, across Scopes 1, 2, and 3 GHG emissions, approximately 40 percent of the assurance performed utilizes ISO 14064-3. Application of other assurance standards are reported to be also consistent across Scopes 1, 2, and 3 GHG emissions: around 10 percent for AICPA, around 10 percent for AccountAbility's AA1000, around 16 percent for IAASB ISAE, and around 30 percent for miscellaneous in-house assurance standards and protocols.[2717] An analysis of S&P 500 firms in 2021 reveals a similar finding with ISO 14064-3 being the most common assurance standard referenced in ESG reporting followed by the IAASB ISAE, which experienced an increase of 41 more references compared to the previous year (i.e., 54 percent increase).[2718]

2713 As discussed in section II.I.5.c, assurance services are services performed in accordance with professional standards that are designed to provide assurance, while in many cases verification services are not designed to provide assurance.

2714 Consistent with rates of voluntary GHG emissions disclosures, the percentages become much smaller when the sample analyzed is expanded to include smaller registrants. The breakdown for LAF, AFs, and NAFs is as follows: 25%, 4%, and 1%, respectively.

2715 *See* Gipper *et al.* (2023).

2716 *See* Center for Audit Quality, *S&P 500 ESG Reporting and Assurance Analysis* (2023), available at https://www.thecaq.org/sp-500-and-esg-reporting (stating, in the context of the study, that the most common standard used by non-accountant providers was ISO 14064-3).

2717 Percentages do not add up to 100% because assurance statements can sometimes reference multiple assurance standards.

2718 Center for Audit Quality, *supra* note 2716.